PSOC™ Edge designing a secure system

Customer training workshop (CTW)

January 2026

public

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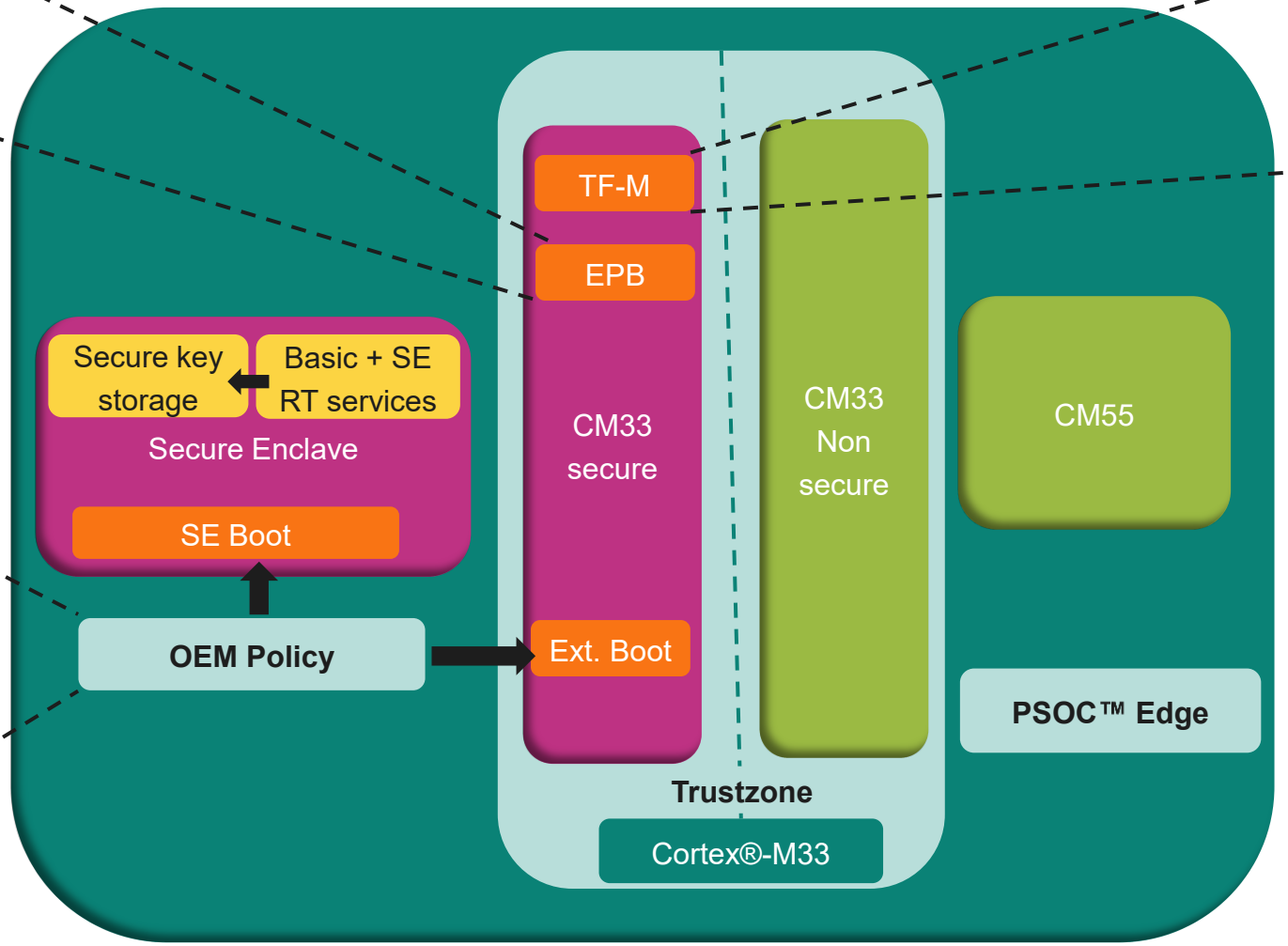
Secure design overview (EPC 4)

Edge Protect Bootloader
- Boot Image validation
- Upgrade image validation
- Encrypted boot
- Downgrade prevention
- Measured boot and sharing

OEM Policy
- Secure boot
- Debug port configuration
- Production LCS
- Re-provisioning policy



Trusted Firmware-M
- CRYPTO services
- Protected storage , ITS
- Attestation services
- Isolation levels
- Application RoT

Legend

Secure

Non-secure

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Secure design overview (EPC 2)

Edge Protect Bootloader
- Boot Image validation
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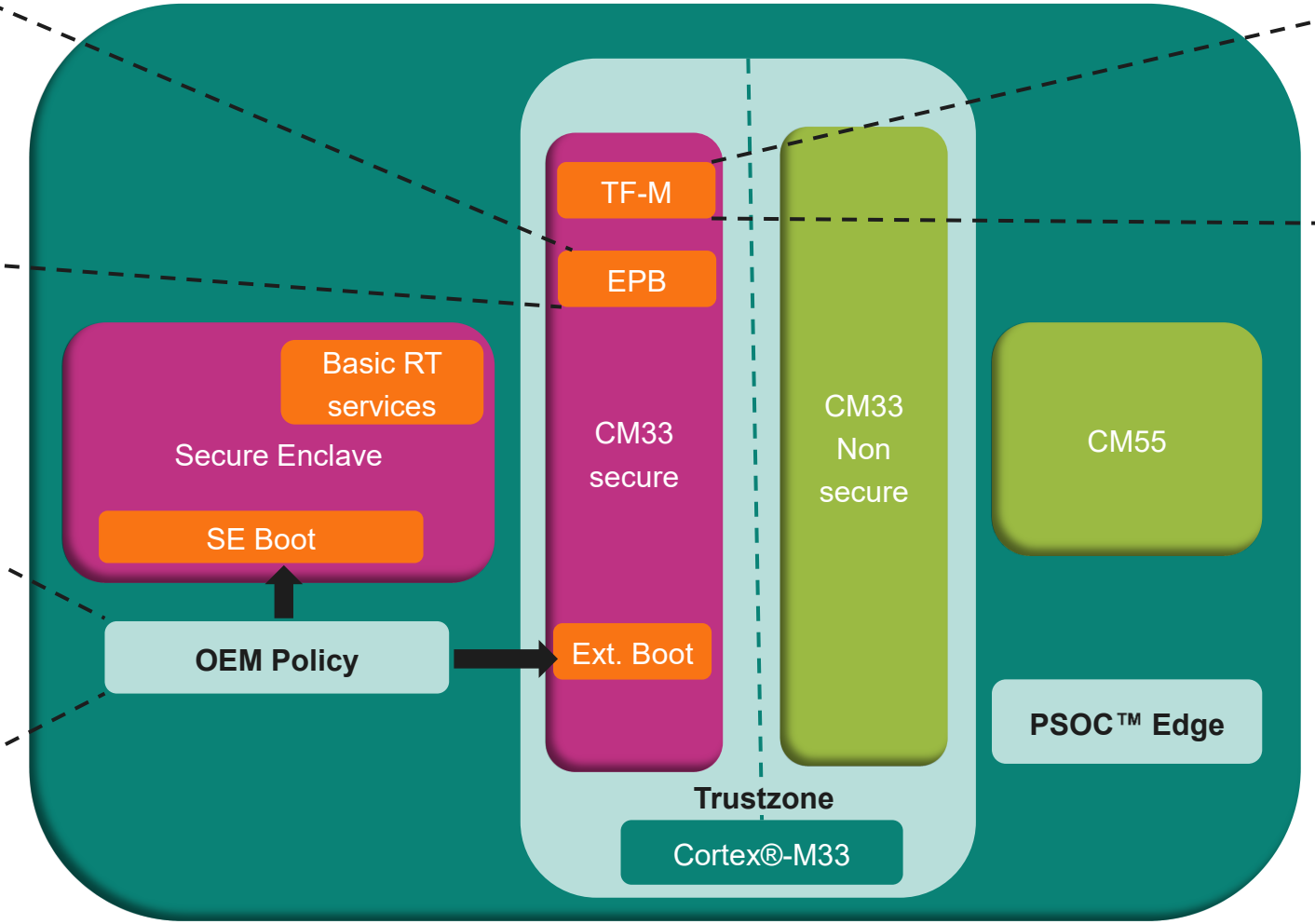

Legend

Secure

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Peripheral and memory protection controller (MPC and PPC)

- PSOC™ Edge memory and peripheral protection can be configured using the peripheral protection controller (PPC) and the memory protection controller (MPC) regions
- The PPC defines the access permissions of a peripheral resource, such as a serial communication block (SCB)
- The MPC is similar, but it applies to memory regions. It is akin to Arm® implementation defined attribution unit (IDAU), but more granular
- Together, these allow for a highly customizable security configuration for a given device
- MPC and PPC regions can be configured using the ModusToolbox™ device configurator

Note: Device configurator protections are applied within the cybsp_init() call from the CM33 secure application

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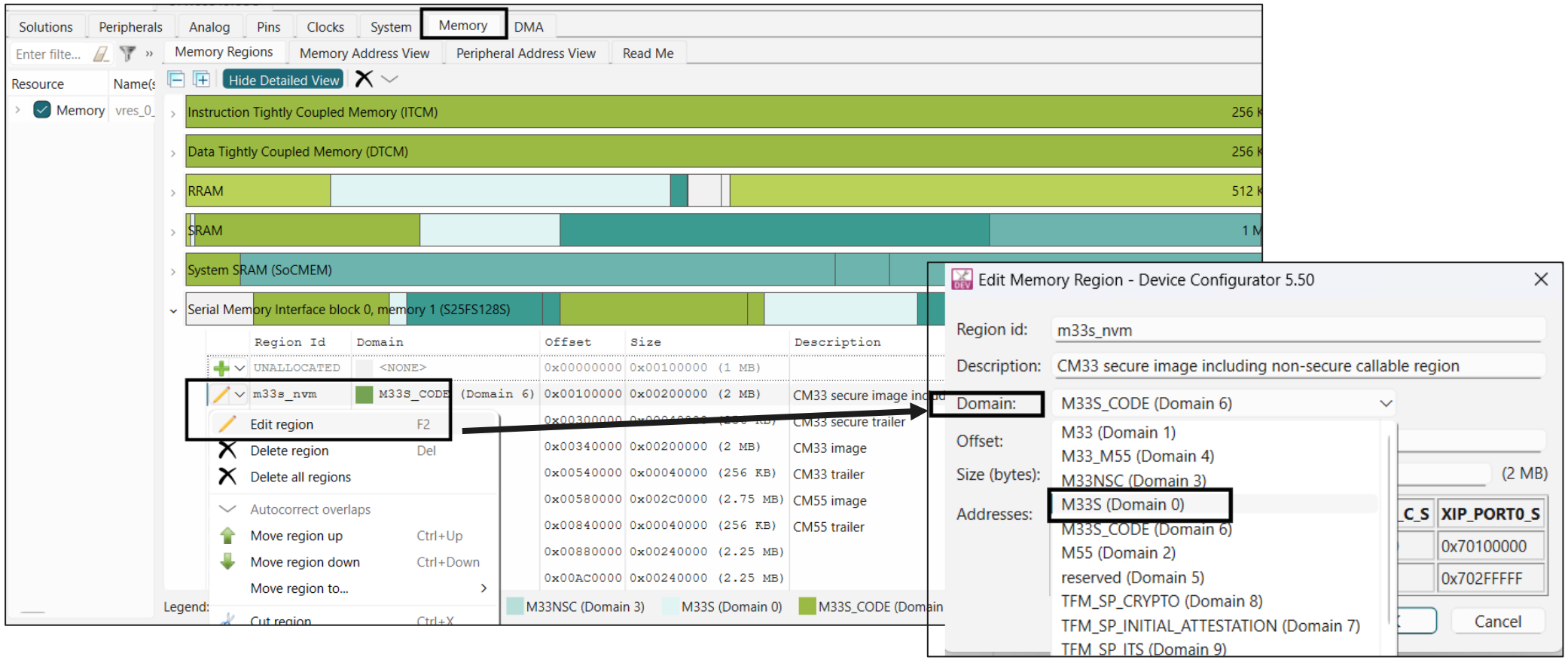
Configure memory protections

– Choose appropriate domains for each memory region



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Domain parameters

PSE846GPS4DBZC4A | CYW55513IUBG

Solutions | Peripherals | Analog | Pins | Clocks | System | Mem

Enter filter text...

Resource	Name(s)
Protection	vres_0_protection_0
Domains	
Domain 0	M33S
Domain 1	M33
Domain 2	M55
Domain 3	M33NSC
Domain 4	M33_M55
Domain 5	reserved
Domain 6	vres_0_protection_0_domain_6
Domain 7	vres_0_protection_0_domain_7
Domain 8	vres_0_protection_0_domain_8
Domain 9	vres_0_protection_0_domain_9
Domain 10	vres_0_protection_0_domain...
Add: Domain 11 (12 of 15)	vres_0_protection_0_domain_11
Memory Protection Controller (MPC)	vres_0_protection_0_mpc_0
Memory Protection Units (MPU)	
Peripheral Protection Controllers (PPC)	
Peri 0	peri_0

Domain 0 (M33S) - Parameters

Enter filter text...

Name	Value
Overview	
Configuration Help	Open Domain D
Top level Protection	
Access	Secure
Peripheral Protection	
Allow peripheral access in unprivileged mode	false
Accessible by:	
PC 2: CM33: Extended Boot, Bootloader, CM33 Secure, CM33 Non-Secure	true
PC 4: Reserved	false
PC 5: Reserved	false
PC 6: CM55 Non-Secure	false
PC 7: Debugger	true
Memory Protection	
PC 2: CM33: Extended Boot, Bootloader, CM33 Secure, CM33 Non-Secure	
Allow reads	true
Allow writes	true
Secure?	true
PC 7: Debugger	
Allow reads	true
Allow writes	true
Secure?	true

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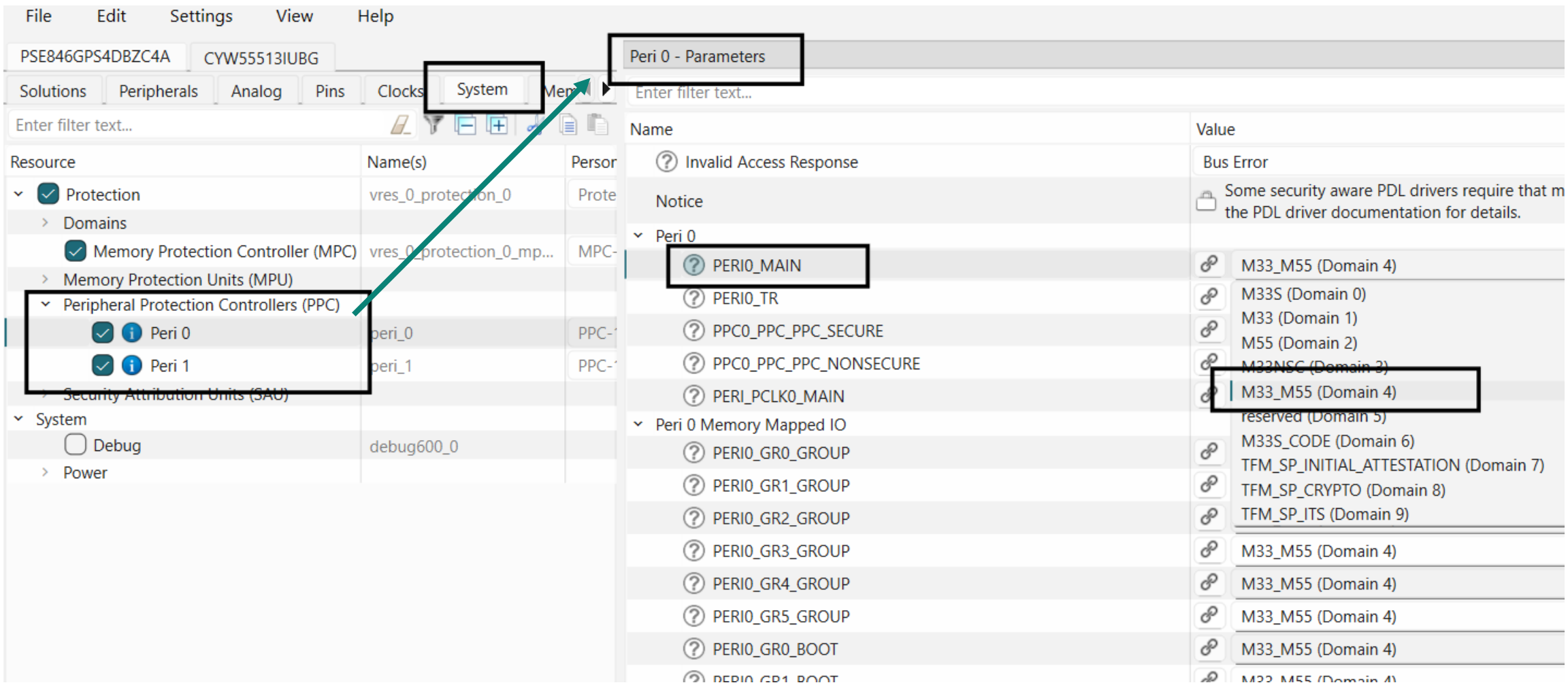
Configure peripheral protections



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ModusToolbox™ secure request framework (SRF)

- The ModusToolbox™ secure request framework (SRF) provides a reusable infrastructure for implementing multiple secure operations via a single non-secure callable (NSC) entry point, which also performs basic argument validation

- It supports calling across the security barrier of a single core, and also calling from a non-TrustZone core to a TrustZone core via an Interprocessor communication (IPC)

- It integrates Arm® Cortex®-M security extensions (CMSE) and Infineon's peripheral protection controller (PPC) and Memory Protection Controller (MPC) regions into a single streamlined process.

Note: See the SRF user guide for more details

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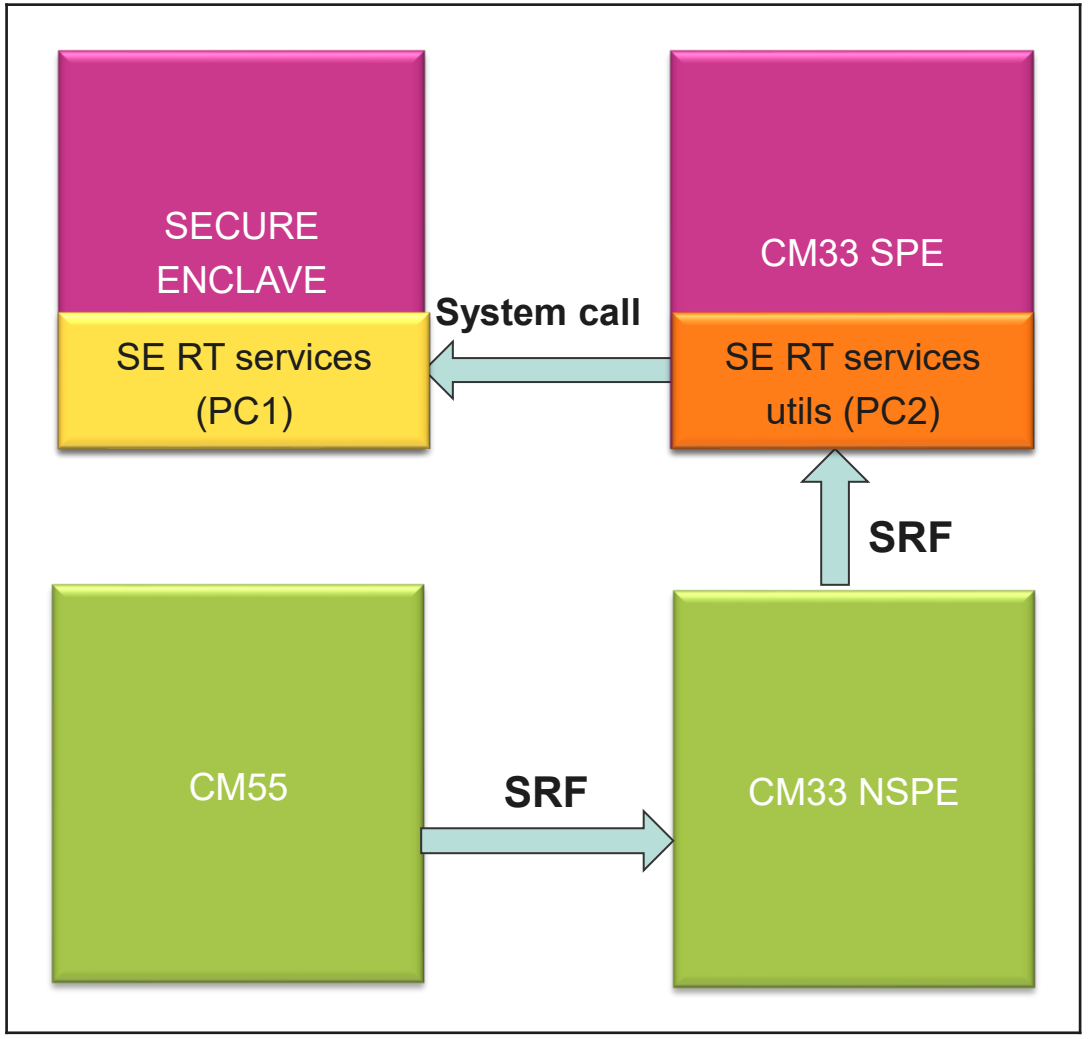
Secure Enclave Runtime (SERT) services (EPC 4)

- Edge Protect Category (EPC) 4 devices provide an additional set of services called secure enclave runtime services (SERT_Services), which include
  - CRYPTO service
  - Key management service
  - Protection service
  - Attestation service
  - System management service

- SERT services utilities library is a set of utility code that provides an interface to SERT services

- SERT services utilities can be accessed from the CM33 SPE application.

Note: Ensure peripherals are configured as secure. See training on "Security tips and tricks" for more details.



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SERT services key storage (EPC 4)

SE RT Services also provide special storage within the Secure Enclave to store persistent keys

User keys can be imported into the Secure Enclave in two ways

1. During runtime using ifx_se_import_key() API.
2. By providing the key in the “user keys” section in the policy file during provisioning

See https://github.com/Infineon/se-rt-services-utils for more details.

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Transition the device to production LCS

- Before the device is shipped out to the field, ensure that the device is in the production LCS
- Ensure that two OEM key pairs are generated and stored securely (recommended to use HSM for key generation and signing)

```
$ edgeprotecttools create-key --key-type ECDSA-P256 --output keys/oem_private_key_0.pem
keys/oem_public_key_0.pem
```

```
$ edgeprotecttools create-key --key-type ECDSA-P256 --output keys/oem_private_key_1.pem
keys/oem_public_key_1.pem
```

- OEM needs to generate the Certificate Signing Request (CSR) with the production type certificate

```
$ edgeprotecttools -t pse8xs2 oem-csr --certificate-name "oem-cert" --oem "Dummy OEM" --
project "Dummy Project" --project-number "1234" --public-key-0 keys/oem_public_key_0.pem
--public-key-1 keys/oem_public_key_1.pem --cert-type production --output packets/apps/
prov_oem/oem_csr_prod.bin --sign-key-0 keys/oem_private_key_0.pem --sign-key-1 keys/
oem_private_key_1.pem
```

- See the application note: Getting started with PSOC™ Edge security for instructions to transition the device to Production LCS

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Re-provisioning

- In the production LCS, the device can only be re-provisioned. Provisioning is not allowed
- The OEM policy fields that need to be altered via re-provisioning must be configured as re-provisionable in the reprovisioning policy object of the OEM policy file. This OEM policy file must be used during the transition to Production LCS
- Only those fields in the OEM policy that were enabled for re-provisioning can be altered later via the re-provisioning process.

Note: If the system access port is closed during the transition to Production LCS, then re-provisioning is not possible through the SWD interface. Alternatively, the user may close the ports using the re-provisioning step.

```
$ edgeprotecttools -t pse8xs2 reprovision-device -p policy/policy_oem_provisioning.json --key
keys/oem_private_key_0.pem
```


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Debug ports configuration

- Configure the debug ports in the OEM policy file and provision the device
- Infineon recommends all the debug ports must be disabled for the final production units.
- Certificate based authenticated debug mechanism is supported
- The device cannot be re-provisioned via SYS-AP once this port has been disabled in the Production LCS.
- By design, the SYS-AP always remains open in the Development LCS to ensure the device can always be provisioned to the default state and does not get bricked due to incorrect policy configurations.

```
"debug": {
  "cm33": {
    "cpu": {
      "description": "CPU access port debug restriction",
      "applicable_conf": "Disabled, Enabled, Allowed",
      "value": "Disabled"
    },
    "allowed": {
    "dbgen": {
    "niden": {
    "spiden": {
    "spniden": {
    "secure": {
  },
  "system": {
    "ap": {
      "description": "System access port (SYS-AP) restriction. The SYS
      "applicable_conf": "Disabled, Enabled, Allowed",
      "value": "Disabled"
    },
    "allowed": {
    "mpc_ppc_enable": {
    "rram_otp_enable": {
    "rram_nvm_enable": {
    "ram_enable": {
    "mmio_enable": {
  },
```


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Revision history

Document revision	Date	Description of change
**	2026-01-12	Initial release.

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